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HACK101: Networks

Introduction to Network Vulnerability Assessments & Penetration Testing



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About Me



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- > Founder of cyber673 community
- > Penetration tester & security assessor
- > HACK101 workshop series organiser

Disclaimer



Educational Purposes Only

Everything taught in this workshop is strictly for educational and authorised testing purposes. Unauthorised testing of any systems, networks, or websites is illegal.

Computer Misuse Act (Brunei)

- > Criminalises unauthorised access to computer systems
- > Penalties include fines and imprisonment
- > Always obtain written authorisation before testing
- > Applies even to "white-hat" or well-intentioned testing

What is HACK101?



A series of introductory workshops covering various types of technical security assessments (VAPT's)

Network

Today!

Web App

Coming soon

Mobile

Coming soon

Active Directory

Coming soon

Thick Client

Coming soon

IoT

Coming soon

Pentesting Lifecycle



1

Recon

Gather information
about the target



2

Vuln Assess

Identify potential
vulnerabilities



3

Exploit

Attempt to
exploit findings



4

Report

Document
findings



5

Cleanup

Remove traces
and restore

The Technical Flow

The VAPT Business Process



01

Get Project

Client engagement, scoping discussions, proposal and SOW

02

Planning & Scope

Confirm scope, IPs, rules of engagement, testing windows

03

Initial Test

Perform the VAPT, document findings as you go

04

Initial Report

Deliver findings report with risk ratings and recommendations

05

Retest

Client remediates; verify fixes in a retest cycle

06

Retest Report

Final report confirming remediation status of each finding

Network VAPT



What is a Network VAPT?

A systematic assessment of network infrastructure to identify vulnerabilities in hosts, services, and configurations that could be exploited by attackers.

Key Areas

- > Host discovery and service enumeration
- > Vulnerability scanning and validation
- > Service configuration review (SSH, SSL/TLS, FTP, etc.)
- > Authentication and access control testing

Nmap: Host Discovery



The essential tool for network scanning and enumeration

```
$ nmap -sn 192.168.1.0/24
```

Ping sweep — discover live hosts

```
$ nmap -sS -p- <target>
```

SYN scan all 65535 ports

```
$ nmap -sV -sC <target>
```

Version detection + default scripts

```
$ nmap -O <target>
```

OS fingerprinting

Nmap: Vulnerability Assessment



Using NSE (Nmap Scripting Engine) for vulnerability detection

```
$ nmap --script vuln <target>
```

Run all vuln category scripts

```
$ nmap --script ssl-enum-ciphers -p 443 <target>
```

Enumerate SSL/TLS cipher suites

```
$ nmap --script ssh2-enum-algos -p 22 <target>
```

Enumerate SSH algorithms

```
$ nmap --script ftp-anon -p 21 <target>
```

Check for anonymous FTP access

Nuclei & Other Tools



Nuclei

Template-based vulnerability scanner

```
$ nuclei -u <target>
```

Run all templates against target

```
$ nuclei -u <target> -t network/
```

Network-specific templates only

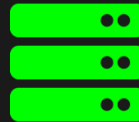
```
$ nuclei -l targets.txt -severity  
critical,high
```

Scan list, filter by severity

Other Useful Tools

- > Nessus / OpenVAS
Full vulnerability scanners
- > testssl.sh
SSL/TLS configuration testing
- > ssh-audit
SSH server configuration audit
- > Nikto
Web server scanner

Finding: Outdated Components



The Issue

Servers and services running outdated software with known CVEs.

Examples:

- > Apache 2.4.29 (known RCE vulns)
- > OpenSSH 7.2 (username enum)
- > nginx 1.14 (header injection)
- > vsftpd 2.3.x (backdoor risk)

Detection

```
$ nmap -sV <target>
```

Identify service versions

```
$ searchsploit apache 2.4.29
```

Search for known exploits

Risk Rating:

High to Critical

Known exploits often publicly available

Finding: Misconfigured Services



Anonymous FTP Access

```
$ ftp <target>
```

Login: anonymous | Password: (blank)

Allows unauthenticated file access, potential data exposure or upload of malware

Default Credentials

- > admin:admin
- > root:toor
- > test:test

Check default creds databases

Open Services

- > Unnecessary ports exposed
- > Management interfaces public
- > Debug endpoints active

Reduce attack surface!

Finding: Insecure Crypto Config



SSH Cipher Issues

Weak algorithms detected:

- > arcfour, arcfour128, arcfour256
- > 3des-cbc, blowfish-cbc
- > hmac-md5, hmac-sha1

```
$ ssh-audit <target>
```

```
$ nmap --script ssh2-enum-algos
```

SSL/TLS Issues

Common problems:

- > SSLv3 / TLS 1.0 / TLS 1.1 enabled
- > Weak ciphers (RC4, DES, NULL)
- > Self-signed / expired certificates
- > Missing HSTS headers

```
$ testssl.sh <target>:443
```

```
$ nmap --script ssl-enum-ciphers
```

Remediation Guide



Outdated Components

- > Patch management policy
- > Automated updates where safe
- > Regular vulnerability scans
- > Subscribe to CVE alerts
- > EOL software migration

Misconfigured Services

- > Disable anonymous FTP
- > Change all default creds
- > Close unnecessary ports
- > Implement firewall rules
- > Principle of least privilege

Insecure Crypto

- > Disable SSLv3, TLS 1.0/1.1
- > Remove weak ciphers
- > Use trusted CA certs
- > Harden sshd_config
- > Enable HSTS headers

CVSS 3.1: Severity Ratings



Common Vulnerability Scoring System — the industry standard for rating vulnerability severity

None 0.0	Low 0.1–3.9	Medium 4.0–6.9	High 7.0–8.9	Critical 9.0–10.0
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Base Score Metrics

Exploitability:

- > Attack Vector (Network/Adjacent/Local/Physical)
- > Attack Complexity (Low/High)
- > Privileges Required (None/Low/High)
- > User Interaction (None/Required)
- > Scope (Unchanged/Changed)

Impact Metrics

CIA Triad:

- > Confidentiality Impact (None/Low/High)
- > Integrity Impact (None/Low/High)
- > Availability Impact (None/Low/High)

Calculator:

nvd.nist.gov/vuln-metrics/cvss/v3-calculator

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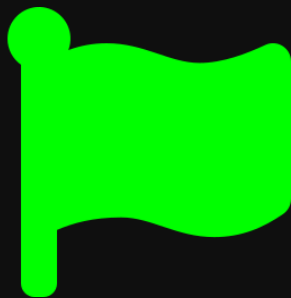


LIVE DEMO

Let's put theory into practice

Scanning the demo server

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CTF CHALLENGE

8 challenges across 2 servers — find the flags!

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